

The Reconstructed Theory Has One Mass: a Machine-Checked Volume-Uniform Spectral Gap for a Concrete Transfer Operator

Lluís Eriksson*

2026-08-04 — complete manuscript; submission is the owner’s decision

Abstract

For the spatial $\mathbb{Z}_2$ (Ising-slice) system in the Dobrushin window $2 \tanh |\beta| + 2 \tanh |\gamma| \leq \alpha < 1$, we machine-check in Lean 4 that the Osterwalder–Schrader (site-form) reconstruction of the transfer operator carries *one* mass: there is a single $m > 0$ such that for *every* spatial extent L the reconstructed (site-form) transfer operator is unitarily conjugate — by the explicit $\sqrt{w}$ boundary dressing — to the symmetrised kernel whose projected norm the Dobrushin corollary bounds by e^{-m} , and every connected correlator of the transported transfer operator decays as $|\text{connCorr } T \Omega v n| \leq \|v\|^2 (e^{-m})^n$ — constant per-observable, rate alone uniform in the volume. All quantifiers live in the Lean statements; the prose here copies the kernel’s cut. To situate the claim in the 2026 landscape of machine-checked constructive field theory: the abstract construction (reflection-positive data $\rightarrow$ physical Hilbert space $\rightarrow$ self-adjoint contractive transfer operator) has been machine-checked, sorry-free, in the Harvard formalization programme, *with the spectral gap carried as an undischarged hypothesis* (`GappedTransfer`); the Osterwalder–Schrader axioms have been machine-checked for the free field; and measure-side Dobrushin clustering has been machine-checked for lattice Yang–Mills. What we add, and what to our knowledge no machine-checked artefact yet contains, is the combination {concrete interacting model, gap *proved* rather than hypothesised, uniform in the spatial volume} on the reconstructed operator — i.e. a concrete discharge of exactly the hypothesis shape the abstract frameworks carry. We do not claim Wightman reconstruction, a continuum limit, $SU(N)$, or any progress on the Clay problem.

1 Introduction and landscape

The Osterwalder–Schrader route builds a physical theory out of Euclidean data: a reflection-positive form gives a Hilbert space, and composition with the time shift gives a self-adjoint transfer operator. For a lattice slice system the construction is elementary — the content is never the construction but the *spectrum*: a spectral gap at every fixed spatial extent L is classical (Perron–Frobenius; in this programme it is the earlier `coupled_gap_all_eigenvalues`), and it degenerates as $L \rightarrow \infty$ unless something uniform survives. What a thermodynamic limit can actually consume is *one* $m > 0$ valid for every L at once.

*With machine-checked artefacts produced on the programme’s Colab plane; toolchain `leanprover/lean4:v4.29.0-rc6`, Mathlib pinned to `0764272048...`. Repository: THE-ERIKSSON-PROGRAMME, branch `d3-closure`, anchor `8127e5b52d75116b19c172ea022e12c1cab0f8c2`.

That uniform input exists in this repository: the Dobrushin corollary `dobrushin_ising_uniform_gap` produces, inside the window $2 \tanh |\beta| + 2 \tanh |\gamma| \leq \alpha < 1$, a single $m > 0$ bounding the projected transfer operator of the tilted kernel at every extent. The present paper’s contribution is the *identification*: the operator that bound governs *is* — unitarily, by the explicit $\sqrt{w}$ boundary dressing, with no spectral-transport theory — the transfer operator forced on the reconstructed (site-form) theory, and the bound transports to volume-uniform clustering of its connected correlators. The theorem is deliberately cheap: the same witnesses, no new estimate. Cheapness is the honesty of the claim — the uniformity is real because it is D-6’s, and the reconstruction adds exactly one commuting square, machine-checked.

1.1 Machine-checked prior work, cited before a referee does

The following table is load-bearing and its rows were collected by a four-modality search on 2026-08-04 (evidence: `docs/OS-R-PRIORITY-COLLATION.md` in the repository).

Artefact	What is machine-checked	What is not
<code>mrdo格拉斯ny/reflection-positivity</code> (sorry-free ~2026-06)	$\text{Positivity} \rightarrow$ Hilbert space (completion/quotient) $\rightarrow$ self-adjoint contraction T ; trace formulas; susceptibility bounds from a gap	the gap itself: carried as the GappedTransfer hypothesis ; no spatial-volume-uniform statement; concrete instance is finite-lattice RP only
<code>OSforGFF</code> (arXiv:2603.15770)	all five GJ/OS axioms for the $d=4$ Gaussian free field, 0 sorries	reconstruction (listed as next milestone); any interacting model
<code>xiyin137/OSreconstruction</code>	Wightman-side GNS lane of the full $\text{OS} \leftrightarrow$ Wightman theorem	the theorem: 53 sorries + 12 axioms at last README snapshot
<code>mrdo格拉斯ny/lgt</code> (complete 2026-05-04)	Chatterjee-style Dobrushin uniqueness for lattice YM, $U(n)$, with exponential clustering whose constants depend only on (n, d, β)	an operator gap; any reconstruction; the statement is per-volume on the measure side
<code>pphi2</code>	transfer-matrix spectral machinery for φ_2^4	rests on 27–29 project axioms
This paper	the $\sqrt{w}$ -dressing unitary; the transported volume-uniform gap; volume-uniform clustering of the reconstructed operator <code>(os_reconstruction_uniform_gap)</code> <code>os_reconstruction_uniform_clustering</code> ; core 8481, oracle 3017, 0 sorry)	Wightman axioms; continuum limit; $SU(N)$; Clay; the raw-measure n -step identification (named next)

2 The two geometries and the forced operator

Everything in this section is prior work of the repository, cited by module (the public trace of this lane is the `ai.viXra 2607.009x` cluster); no new mathematics appears here. The slice

system carries two pairings on complex boundary vectors: the *site* form, which weighs by $1/w$ (`siteForm`, module `SpatialReconstruction`), and the *bond* form carrying the decoupled kernel `spatialKernel` β (`bondForm`). The transfer operator is *forced*: $T = D_w K$ is the unique operator with $\text{site}(u, Tv) = \text{bond}(u, v)$ (`transferOp_unique`), self-adjoint for the site form at every β (`transferOp_selfAdjoint`) and positive for $\beta \geq 0$; reflection positivity of the underlying measure is the lane's earlier bond/site pair (modules `SpatialReflection`, `SpatialOS`). On the Dobrushin side, the same repository carries `symWeighted` $= \sqrt{w} K \sqrt{w}$ (module `PerronGap`), its normalisation `tiltKernel` $= \text{symWeighted}/\lambda$ (module `DobrushinTilt`), the Euclidean packaging `opOf`, `vacOf`, `vacuumTransfer_opOf` (module `DobrushinTransport`), and the corollary `dobrushin_ising_uniform_gap` (module `DobrushinCorollary`). The seam fact this paper depends on — one `spatialKernel` declaration serving both geometries — is enforced by the elaborator in the module of Section 3.

3 The dressing unitary

All three statements are green on the plane (module `YangMills/OS/OSReconstructionUniform.lean`, sha256 5445429f . . . , five-pass ladder, `ELAB_EXIT 0`). Write $Qu := \sqrt{w} u$ for the boundary dressing of a real Euclidean vector (`qEmbed`); the site form weighs by $1/w$.

Theorem 3.1 (`A1, siteForm_qEmbed`). *For $w > 0$ and real u, v : $\text{siteForm } w (Qu) (Qv) = \sum_{\sigma} u_{\sigma} v_{\sigma}$. Per site the cancellation is elementary: $\sqrt{w} u \cdot \sqrt{w} v \cdot (1/w) = uv$.*

Theorem 3.2 (`A2, transferOp_qEmbed`). *$T(Qu) = Q(\text{act}(\text{symWeighted } w \beta) u)$ pointwise: the forced transfer operator on a dressed vector is the dressing of the symmetrised kernel's action.*

Theorem 3.3 (`A3, transferOp_iterate_qEmbed`). *$T^{[n]}(Qu) = Q((\text{act } S)^{[n]} u)$ — the shape the in-tree measure bridge `gibbsPathSum_eq_iterate` consumes.*

4 The transported volume-uniform gap

Theorem 4.1 (`os_reconstruction_uniform_gap`). *In the window $2 \tanh |\beta| + 2 \tanh |\gamma| \leq \alpha < 1$ there is $m > 0$ such that for every L there are $\lambda > 0$ and $\Omega > 0$ with the fixed-point equation $\sum_{\tau} \text{tiltKernel}(\text{sliceW } \gamma L) \beta \lambda \sigma \tau \Omega_{\tau} = \Omega_{\sigma}$, the projected bound $\|\text{projectedTransfer} \dots\| \leq e^{-m}$, and the A2 identity at $w := \text{sliceW } \gamma L$.*

The proof is an obtain on `dobrushin_ising_uniform_gap` followed by A2: the same witnesses, no new analytic content — which is precisely the point. The uniformity is real because it is D-6's; the reconstruction contributes the identification, not a new estimate.

5 Identification against the measure

Green at pass 5 (operator side):

Theorem 5.1 (`B0b, act_symWeighted_eq_smul_act_tilt`). *$\text{act } S u = \lambda \cdot \text{act}(\text{tilt}) u$ for $\lambda \neq 0$.*

Theorem 5.2 (`B0a, act_iterate_eq_opOf_pow`). *$(\text{act } M)^{[n]} u = (\text{opOf } M)^n (\text{toLp } u)$ pointwise.*

Theorem 5.3 (`os_reconstruction_uniform_clustering`). *In the window: ONE $m > 0$ such that for EVERY spatial extent L , every connected correlator of the transported transfer operator obeys $|\text{connCorr } T \Omega v n| \leq \|v\|^2 \cdot (e^{-m})^n$. The constant is per-observable; the rate alone is uniform in L .*

The packaging is by citation: `vacuumTransfer_op0f + tiltKernel_symm + clustering_of_gap`, all in-tree. One step of the campaign charter remains open, and we name it rather than blur it: the *raw-measure* n -step identification — the statement that the unnormalised Gibbs two-point sums (`gibbsPathSum`, whose operator bridge `gibbsPathSum_eq_iterate` is already in the tree) equal λ^n times matrix elements of the operator powers through B0a/B0b and A3, together with the division-safe six-term connected bound against the partition function. Its bricks are designed and its judge identities (G5 of `judge_os_uniform.py`) are verified numerically at $n \leq 4$; it is fabrication, not research. Nothing in Sections 3–5 depends on it: the clustering theorem above is a statement about the reconstructed operator and stands on its own.

6 Reproduction

All numbers below were read from committed run artefacts (ledger Addenda 616–618); predictions were committed *before* their measurements.

Quantity	Predicted	Measured
core jobs, unwired	8480 (ledger baseline)	8480
core jobs, wired	$8480 + 1$	8481
oracle reports, wired	$3010 + 7$	3017
<code>sorryAx</code> occurrences	0	0
OS-R endpoint axiom sets	standard triple	exactly <code>[propext, Classical.choice, Quot.sound] × 7</code>

Module: `YangMills/OS/OSReconstructionUniform.lean`, sha256 5445429f7736...30cad1 (12,347 bytes), five elaboration passes on a fresh high-RAM Colab runtime (toolchain v4.29.0-rc6, Mathlib pinned); wired at anchor 8127e5b52d75116b19c172ea022e12c1cab0f8c2. The count prediction $+1/ + 7$ held exactly — the ninth consecutive exact count prediction of the programme’s D/OS lane.

Fresh-clone witness (ledger Addendum 619): a brand-new clone at the anchor, toolchain and Mathlib cache from scratch, reproduced — judges 110/110 in normal and -0; core Build completed successfully (8481 jobs); oracle REPORTS 3017, SORRYAX 0, zero reports naming any axiom outside `[propext, Classical.choice, Quot.sound]` (a continuation-joining checker; declarations depending on a strict *subset* of the triple occur and are sound); module file sha256 5445429f...30cad1 bit-identical to the committed tree. The run cell is committed (`scripts/osr1_pass7_freshc` for any further independent reproduction.

7 What is not claimed

No Wightman axioms, no $\text{OS} \rightarrow \text{Wightman}$ analytic continuation, no continuum limit, no $SU(N)$, no statement about the Clay problem. The window is the Dobrushin window; outside it the programme’s own kill-test (repository, Addendum 610) shows the standard-cone Birkhoff route is projectively blind to the site weight, which is why no claim beyond the window appears here.